

Bioinformatics Internship Report

Task 1: DNA/Protein Sequence Analysis (BLAST Analysis)

1. Introduction

The primary objective of this task was to perform a sequence homology search using the Basic Local Alignment Search Tool (BLAST). By comparing a query protein sequence against a non-redundant database, we aim to identify homologous sequences, conserved domains, and evolutionary relationships.

2. Methodology

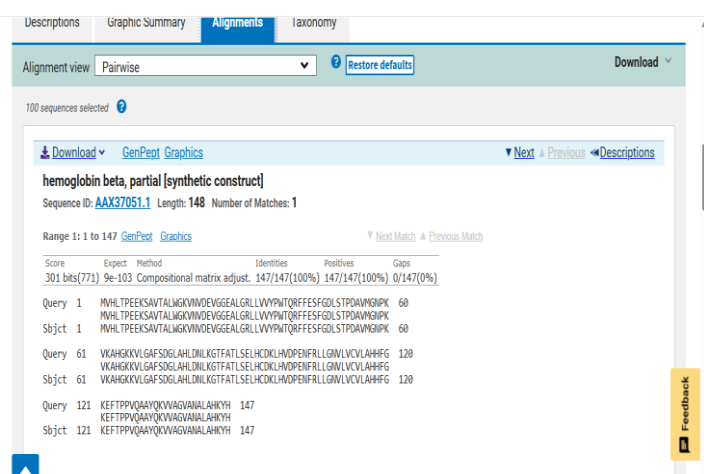
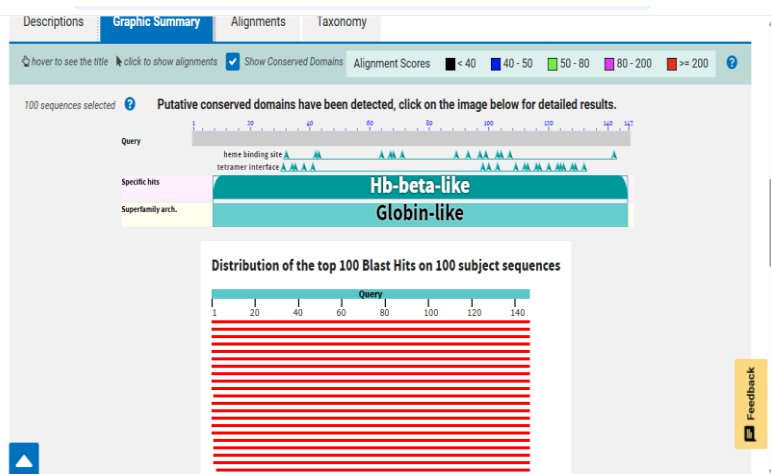
Program Used	BLASTp (Protein-Protein BLAST)
Database	Non-redundant protein sequences (nr)
Query ID	lcl Query_3487745
Molecule Type	Amino Acid

3. BLAST Results Analysis

3.1. Graphic Summary & Conserved Domains

The graphic summary reveals that the query sequence belongs to the **Globin-like superfamily**. Specifically, a high-confidence match for the **Hb-beta-like** domain was detected. Key functional features identified include:

- Heme binding sites:** Multiple conserved residues responsible for binding the prosthetic heme group.
- Tetramer interface:** Residues involved in the interaction between different subunits of the hemoglobin molecule.



Descriptions	Graphic Summary	Alignments	Taxonomy					
Sequences producing significant alignments								
Download ▼ Select columns ▼ Show 100 ?								
☒ select all 100 sequences selected								
GenPeptGraphicsDistance tree of resultsMultiple alignmentMSA Viewer								
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession
☒ hemoglobin beta, partial [synthetic construct]	synthetic const...	301	301	100%	9e-103	100.00%	148	AAX37051.1
☒ hemoglobin beta, partial [synthetic construct]	synthetic const...	301	301	100%	1e-102	100.00%	148	AAX28657.1
☒ hemoglobin subunit beta [Homo sapiens]	Homo sapiens	301	301	100%	1e-102	100.00%	147	NP_000509.1
☒ hemoglobin subunit beta [Gorilla gorilla gorilla]	Gorilla gorilla...	300	300	100%	4e-102	99.32%	147	XP_018891709.1
☒ beta globin chain variant [Homo sapiens]	Homo sapiens	299	299	100%	5e-102	99.32%	147	AAN84548.1
☒ beta-globin [Homo sapiens]	Homo sapiens	299	299	100%	5e-102	99.32%	147	ACU56984.1
☒ beta globin [Homo sapiens]	Homo sapiens	299	299	100%	5e-102	99.32%	147	AAZ39780.1
☒ hemoglobin beta chain [Homo sapiens]	Homo sapiens	299	299	100%	6e-102	99.32%	147	AAD19696.1
☒ HBB, partial [synthetic construct]	synthetic const...	299	299	100%	6e-102	99.32%	147	AKI70610.1
☒ HBB, partial [synthetic construct]	synthetic const...	299	299	100%	9e-102	99.32%	147	AKI70611.1
☒ Chain B, HEMOGLOBIN (BETA CHAIN) [Homo sapiens]	Homo sapiens	298	298	99%	9e-102	100.00%	146	1A3N_B
☒ hemoglobin subunit beta [Homo sapiens]	Homo sapiens	298	298	100%	1e-101	99.32%	147	QBH498124.1

3.2. Significant Alignments (Top Hits)

The distribution of the top 100 hits shows a uniform red color across the entire query length (Score ≥ 200), indicating extremely high homology. The top sequences producing significant alignments include:

- **Hemoglobin subunit beta [Homo sapiens]:** 100% Query Coverage, 100% Identity, E-value: 1e-102.
- **Hemoglobin subunit beta [Gorilla gorilla gorilla]:** 100% Query Coverage, 99.32% Identity, E-value: 4e-102.

4. Results Interpretation

Statistical Significance: The E-values (ranging from 9e-103 to 1e-101) are exceptionally low, indicating that these matches are biologically significant and not due to random chance.

Sequence Identity: A 100% identity match with Human Hemoglobin Beta confirms the query sequence's origin. The 99.32% identity with Gorilla Hemoglobin further validates the highly conserved nature of this protein across primates.

5. Conclusion

The BLASTp analysis successfully identified the query sequence as Hemoglobin Subunit Beta. The high degree of conservation in functional domains (heme binding and tetramer interface) underscores its critical role in oxygen transport. This data serves as a foundation for subsequent tasks, including Multiple Sequence Alignment and 3D Structure Prediction.
